

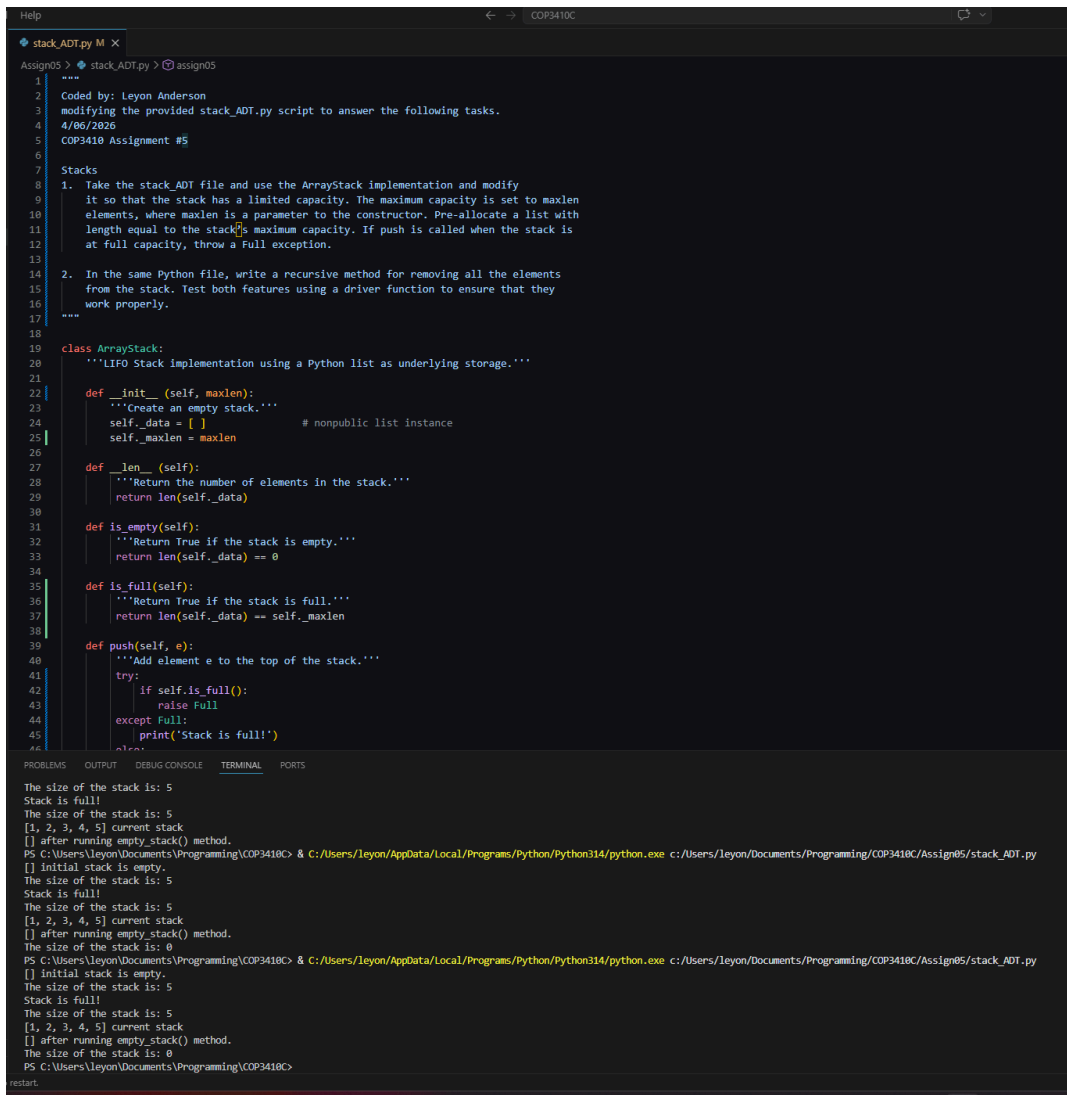
COP3410 Florida Atlantic University Assignment 5

100 points

Stacks

1. Take the stack_ADT file and use the ArrayStack implementation and modify it so that the stack has a limited capacity. The maximum capacity is set to maxlen elements, where maxlen is a parameter to the constructor. Pre-allocate a list with length equal to the stack's maximum capacity. If push is called when the stack is at full capacity, throw a Full exception.

2. In the same Python file, write a recursive method for removing all the elements from the stack. Test both features using a driver function to ensure that they work properly.



```
Help
stack_ADT.py M X
Assign05 > stack_ADT.py > assign05
1
2 Coded by: Leyon Anderson
3 modifying the provided stack_ADT.py script to answer the following tasks.
4 4/06/2026
5 COP3410 Assignment #5
6
7 Stacks
8 1. Take the stack_ADT file and use the ArrayStack implementation and modify
9 it so that the stack has a limited capacity. The maximum capacity is set to maxlen
10 elements, where maxlen is a parameter to the constructor. Pre-allocate a list with
11 length equal to the stack's maximum capacity. If push is called when the stack is
12 at full capacity, throw a Full exception.
13
14 2. In the same Python file, write a recursive method for removing all the elements
15 from the stack. Test both features using a driver function to ensure that they
16 work properly.
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19
20 class ArrayStack:
21     '''LIFO Stack implementation using a Python list as underlying storage.'''
22
23     def __init__(self, maxlen):
24         '''Create an empty stack.'''
25         self._data = [ ] # nonpublic list instance
26         self._maxlen = maxlen
27
28     def __len__(self):
29         '''Return the number of elements in the stack.'''
30         return len(self._data)
31
32     def is_empty(self):
33         '''Return True if the stack is empty.'''
34         return len(self._data) == 0
35
36     def is_full(self):
37         '''Return True if the stack is full.'''
38         return len(self._data) == self._maxlen
39
40     def push(self, e):
41         '''Add element e to the top of the stack.'''
42         try:
43             if self.is_full():
44                 raise Full
45         except Full:
46             print('Stack is full!')
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Queues

3. What values are returned during the following sequence of queue operations, if executed on an initially empty queue? enqueue(5), enqueue(3), dequeue(), enqueue(2), enqueue(8), dequeue(), dequeue(), enqueue(9), enqueue(1), dequeue(), enqueue(7), enqueue(6), dequeue(), dequeue(), enqueue(4), dequeue(), dequeue(). Use the Queue.py file to show the queue contents after every operation.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
[None, None, None, None, None, None, None, None, None, None] initially empty!
[None, None, None, None, None, None, None, None, None, None]
PS C:\Users\leyon\Documents\Programming\COP3418C> & C:/Users/leyon/AppData/Local/Programs/Python/Python314/python.exe c:/Users/leyon/Documents/Programming/COP3418C/Assign85/Queue_ADT.py
[None, None, None, None, None, None, None, None, None, None] initially empty!
[5, None, None, None, None, None, None, None, None, None]
[5, 3, None, None, None, None, None, None, None, None]
[None, 3, None, None, None, None, None, None, None, None]
[None, 3, 2, None, None, None, None, None, None, None]
[None, 3, 2, 8, None, None, None, None, None, None]
[None, None, 2, 8, None, None, None, None, None, None]
[None, None, None, 8, None, None, None, None, None, None]
[None, None, None, 8, 9, None, None, None, None, None]
[None, None, None, 8, 9, 1, None, None, None, None]
[None, None, None, None, 9, 1, None, None, None, None]
[None, None, None, None, 9, 1, 7, None, None, None]
[None, None, None, None, 9, 1, 7, 6, None, None]
[None, None, None, None, None, 1, 7, 6, None, None]
[None, None, None, None, None, None, 7, 6, None, None]
[None, None, None, None, None, None, 7, 6, 4, None]
[None, None, None, None, None, None, None, 6, 4, None]
[None, None, None, None, None, None, None, None, 4, None]
[None, None, None, None, None, None, None, None, None, None] the result
PS C:\Users\leyon\Documents\Programming\COP3418C>
```

4. Suppose an initially empty queue Q has executed a total of 32 enqueue operations, 10 first operations, and 15 dequeue operations, 5 of which raised Empty errors that were caught and ignored. What is the current size of Q? Add your answer to the end of the python file.

```
Queue_ADT.py M
Assign05 > Queue_ADT.py > ...
104     print(A)
105     A.dequeue()
106     print(A)
107     A.enqueue(9)
108     print(A)
109     A.enqueue(1)
110     print(A)
111     A.dequeue()
112     print(A)
113     A.enqueue(7)
114     print(A)
115     A.enqueue(6)
116     print(A)
117     A.dequeue()
118     print(A)
119     A.dequeue()
120     print(A)
121     A.enqueue(4)
122     print(A)
123     A.dequeue()
124     print(A)
125     A.dequeue()
126     print(f'{A} the result')
127
128     """
129     4. Suppose an initially empty queue Q has executed a total of 32 enqueue operations,
130     10 first operations, and 15 dequeue operations, 5 of which raised Empty errors
131     that were caught and ignored. What is the current size of Q? Add your answer to
132     the end of the python file.
133
134     This size of Q should be 32 - 10 = 22, assuming 5 failed dequeue operations. First
135     operations would not impact size.
136     """
```